Pion Current Background-Response Diagnostics

Linearized Current Response and Current-Current Response Propagators

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May 21, 2026

Contents

1 Purpose

This note documents the pion current background-response diagnostics implemented in

`pyquda_measurement_utils/pion_current_background_response_vibe_develop.py`

and

`application/EMFF_pion_background_response/perlmutter/`.

The goal is to study background-field/Feynman-Hellmann routes to pion current matrix elements while *not* modifying the QUDA Dirac operator and while *not* performing inversions with a finite external field $D + \lambda\mathcal{O}$. Instead, the current implementation constructs linearized response propagators using ordinary D^{-1} inversions.

This is not yet the full finite-λ background-field method. It is the linearized method at $\lambda = 0$, which is mathematically equivalent to a summed current insertion.

The pion electromagnetic form-factor (EMFF) response is the first concrete application, using the temporal vector current V_4. The same source module also contains a minimal second-order current-current response building block for future hadronic-tensor or Compton-amplitude style diagnostics.

2 Two Related But Distinct Methods

2.1 Finite-λ background-field method

The most literal background-field method modifies the action or Dirac operator,

$$D_\lambda = D + \lambda\mathcal{O}, \tag{1}$$

then computes ordinary two-point functions in this modified theory,

$$C_\lambda(t) = \langle O_\pi(t) O_\pi^\dagger(0)\rangle_\lambda . \tag{2}$$

At large Euclidean time,

$$C_\lambda(t) = Z_\lambda e^{-E_\lambda t} + \cdots . \tag{3}$$

The desired matrix element is obtained from the energy response,

$$\left.\frac{\partial E_\lambda}{\partial \lambda}\right|_{\lambda=0} . \tag{4}$$

This is the usual "extract the energy shift" picture of the background-field method.

2.2 Linearized response at $\lambda = 0$

The method currently implemented in this repository does not invert $D + \lambda\mathcal{O}$. Instead, it expands the propagator,

$$S_\lambda = (D + \lambda\mathcal{O})^{-1} = S - \lambda S\mathcal{O}S + O(\lambda^2). \tag{5}$$

Thus,

$$\left.\frac{\partial S_\lambda}{\partial \lambda}\right|_{\lambda=0} = -S\mathcal{O}S. \tag{6}$$

The code constructs the object

$$S_{\mathrm{resp}} = S\mathcal{O}S = D^{-1}\mathcal{O}S, \tag{7}$$

using ordinary inversions. Therefore, relative to the derivative in Eq. (6), the physical finite-difference sign is

$$\left.\frac{\partial S_\lambda}{\partial \lambda}\right|_{\lambda=0} = -S_{\mathrm{resp}}. \tag{8}$$

The positive-sign S_{resp} convention is useful because it can be compared directly against the explicit summed three-point function.

2.3 Why the linear response looks like a summed three-point function

Expanding the two-point function in the external source gives

$$\left.\frac{\partial C_\lambda(t)}{\partial \lambda}\right|_{\lambda=0} = -\sum_\tau \langle O_\pi(t)\,\mathcal{O}(\tau)\,O_\pi^\dagger(0)\rangle\,, \tag{9}$$

up to the sign convention chosen in the action perturbation. In the code-sign convention of Eq. (7), the response contraction is compared to

$$\sum_\tau C_3(t,\tau). \tag{10}$$

This explains why a linearized background-response calculation has analysis features similar to a summed three-point function: it is the derivative $\partial C_\lambda/\partial\lambda|_0$, not a finite-$\lambda$ energy fit.

3 Pion Current Kinematics

For the pion EMFF application, the current matrix element is defined by

$$\langle \pi(p_f)|J_\mu(0)|\pi(p_i)\rangle = (p_i+p_f)_\mu F_\pi(Q^2), \qquad q = p_f - p_i. \tag{11}$$

The lattice implementation uses

$$\mathbf{p}_i = \mathbf{p}_f - \mathbf{q}_{\text{ext}}. \tag{12}$$

The explicit three-point correlator is

$$C_3^\Gamma(\mathbf{q},\tau;\mathbf{p}_f,t_{\text{sep}}) = \sum_{\mathbf{x}} e^{-i\mathbf{q}\cdot(\mathbf{x}-\mathbf{x}_0)}\,\mathrm{tr}\left[S_{\text{seq,anti}}(\mathbf{x},\tau;\mathbf{p}_f,t_{\text{sep}})\Gamma S_q(\mathbf{x},\tau;\mathbf{x}_0,0)\Gamma_{\text{src}}\right], \tag{13}$$

where the fixed-sink sequential propagator has already absorbed the sink momentum $\mathbf{p}_f$, the sink time t_{sep}, and the sink interpolator Γ_{sink}. The standard EMFF choice is the temporal vector current,

$$\Gamma = \gamma_T, \tag{14}$$

labeled `T` in the code.

4 Current-Inserted Source

For a local current with momentum transfer $\mathbf{q}$,

$$\mathcal{O}_{\Gamma,\mathbf{q}}(x) = \Gamma\,\Phi_{\mathbf{q}}(x-\mathbf{x}_0), \tag{15}$$

where $\Phi_{\mathbf{q}}$ is produced by `MomentumPhase` using the same phase convention as the ordinary pion current three-point code.

Given an ordinary source-to-current propagator

$$S(x,x_0), \tag{16}$$

the all-time current-inserted source is

$$\eta_{\Gamma,\mathbf{q}}(x) = \Gamma\,\Phi_{\mathbf{q}}(x-\mathbf{x}_0)\,S(x,x_0). \tag{17}$$

The response propagator is then obtained by a standard inversion,

$$S_{\text{resp}}(y,x_0;\Gamma,\mathbf{q}) = \sum_x S(y,x)\,\Gamma\,\Phi_{\mathbf{q}}(x-\mathbf{x}_0)\,S(x,x_0). \tag{18}$$

In code, Eq. (17) is built by

```
build_local_current_inserted_source(...)
```

and Eq. (18) by

```
invert_local_current_response_propagator(...).
```

5 Response Correlator

The response propagator can be contracted like the forward quark line in a pion two-point function. With an ordinary antiquark line S_{anti},

$$C_{\mathrm{resp}}(t;\mathbf{p}_f,\mathbf{q}) = \sum_{\mathbf{y}} e^{-i\mathbf{p}_f\cdot(\mathbf{y}-\mathbf{x}_0)}\,\mathrm{tr}\left[S_{\mathrm{anti}}(\mathbf{y},t;\mathbf{x}_0,0)\,\Gamma_{\mathrm{sink}}\,S_{\mathrm{resp}}(\mathbf{y},t;\mathbf{x}_0,0;\Gamma,\mathbf{q})\,\Gamma_{\mathrm{src}}\right]. \tag{19}$$

For the code-sign convention,

$$C_{\mathrm{resp}}(t_{\mathrm{sep}};\mathbf{p}_f,\mathbf{q}) = \sum_{\tau} C_3^{\Gamma}(\mathbf{q},\tau;\mathbf{p}_f,t_{\mathrm{sep}}), \tag{20}$$

provided the same current gamma, source/sink interpolators, momentum phases, and smearing conventions are used.

The first validated case is

$$\mathbf{p}_f=\mathbf{0},\qquad \mathbf{q}=\mathbf{0},\qquad \mathbf{p}_i=\mathbf{0},\qquad \Gamma=\gamma_T,\qquad \Gamma_{\mathrm{src}}=\Gamma_{\mathrm{sink}}=\gamma_5. \tag{21}$$

On the S8T32 test gauge, the diagnostic found

$$\sum_{\tau} C_3(\tau) = 327.27206385480486 + 0.1342484007638715\,i, \tag{22}$$

$$C_{\mathrm{resp}} = 327.27206385480486 + 0.13424840076386432\,i, \tag{23}$$

with relative difference

$$2.2\times 10^{-17}. \tag{24}$$

6 Energy-Shift Interpretation

Although Eq. (20) looks like a summed three-point function, it is still directly related to the background-field energy shift. Suppose

$$C_\lambda(t) = Z_\lambda e^{-E_\lambda t}. \tag{25}$$

Expanding around $\lambda = 0$,

$$Z_\lambda = Z_0 + \lambda Z_0' + O(\lambda^2), \tag{26}$$

$$E_\lambda = E_0 + \lambda E_0' + O(\lambda^2), \tag{27}$$

gives

$$C_\lambda(t) = C_0(t)\left[1+\lambda\left(\frac{Z_0'}{Z_0} - tE_0'\right)\right] + O(\lambda^2). \tag{28}$$

Therefore

$$\frac{1}{C_0(t)}\left.\frac{\partial C_\lambda(t)}{\partial\lambda}\right|_{\lambda=0} = \frac{Z_0'}{Z_0} - tE_0'. \tag{29}$$

Thus, in the linear-response method, the energy-shift derivative is extracted from the slope of a response ratio, not from separate finite-λ energy fits. This is the same Feynman-Hellmann information, but represented at $\lambda = 0$.

7 Summed-Ratio Object

Define

$$R_{\mathrm{sum}}(t_{\mathrm{sep}}) = \frac{\sum_\tau C_3(t_{\mathrm{sep}}, \tau)}{C_2(t_{\mathrm{sep}})} = \frac{C_{\mathrm{resp}}(t_{\mathrm{sep}})}{C_2(t_{\mathrm{sep}})}. \tag{30}$$

At large t_{sep},

$$R_{\mathrm{sum}}(t_{\mathrm{sep}}) = A - t_{\mathrm{sep}} E_0' + \text{excited-state corrections}, \tag{31}$$

where A absorbs overlap-factor response and contact-term effects.

For the finite-λ sign convention $D_\lambda = D + \lambda\mathcal{O}$, the derivative of the propagator has the sign in Eq. (6). The diagnostic code stores both

```
response_sign = +1
```

for direct comparison to the explicit summed three-point function, and

```
finite_difference_derivative_sign = -1
```

for the corresponding finite-field derivative convention.

8 Restricted Insertion-Time Windows

The full background-field energy-shift picture corresponds to an all-time insertion. However, in the linearized response implementation one may restrict the insertion time,

$$S_{\mathrm{resp}}^{W_\tau} = \sum_{\tau \in W_\tau} D^{-1} \left[\delta_{t,\tau} \, \Gamma \Phi_{\mathbf{q}}(x - \mathbf{x}_0) S(x, x_0) \right]. \tag{32}$$

This produces

$$C_{\mathrm{resp}}^{W_\tau}(t_{\mathrm{sep}}) = \sum_{\tau \in W_\tau} C_3(t_{\mathrm{sep}}, \tau). \tag{33}$$

Useful choices include

$$W_\tau = \text{all time slices}, \tag{34}$$
$$W_\tau = \{0, \ldots, t_{\mathrm{sep}}\}, \tag{35}$$
$$W_\tau = \{1, \ldots, t_{\mathrm{sep}} - 1\}, \tag{36}$$
$$W_\tau = \{\tau_{\min}, \ldots, t_{\mathrm{sep}} - \tau_{\min}\}. \tag{37}$$

The restricted windows are not the pure finite-field energy-shift method. They are analysis tools for the linearized method, useful for:

- removing source/sink contact terms,
- comparing with explicit three-point functions in the same time window,
- studying excited-state contamination,
- checking the stability of summed-ratio slopes.

9 Current Implementation

9.1 Source helper

The helper

```
build_local_current_inserted_source(prop_forward, phase_q, current_gamma, tau)
```

constructs

$$\eta(x) = \Gamma_{\text{cur}} \Phi_{\mathbf{q}}(x - \mathbf{x}_0) S(x, x_0). \tag{38}$$

If `tau` is provided, the source is masked to that global insertion time. If `tau` is omitted, all time slices are included.

9.2 Response inversion

The helper

```
invert_local_current_response_propagator(...)
```

performs the ordinary D^{-1} inversion of the current-inserted source. If a list of times `tau_list` is given, the code first builds the summed time-window source and then performs one inversion. This deliberately avoids storing per-τ response propagators.

9.3 Response contraction

The helper

```
contract_response_pion_2pt(...)
```

contracts the response propagator with the ordinary antiquark line using the same pion two-point contraction infrastructure as the rest of the pion code.

9.4 Current-current response

The same helper module also contains a minimal nested current-current response prototype,

```
invert_current_current_response_propagator(...).
```

For two local currents O_1 and O_2, it constructs

$$S^{(2)}_{\text{resp}} = D^{-1} O_2 D^{-1} O_1 S. \tag{39}$$

With

$$O_i(x) = \Gamma_i \Phi_{\mathbf{q}_i}(x - \mathbf{x}_0), \tag{40}$$

this corresponds to the ordered two-current insertion

$$S^{(2)}_{21} = S O_2 S O_1 S, \tag{41}$$

up to the sign convention chosen for finite-λ derivatives. The associated pion two-point-like response contraction is

$$C^{(2)}_{\text{resp}}(t_{\text{sep}}) = \sum_{\tau_2 \in W_2} \sum_{\tau_1 \in W_1} C_{JJ}(t_{\text{sep}}, \tau_2, \tau_1), \tag{42}$$

where C_{JJ} is the pion correlator with two local current insertions on the active quark line. The total momentum injected by the two currents is

$$\mathbf{q}_{\text{tot}} = \mathbf{q}_1 + \mathbf{q}_2, \qquad \mathbf{p}_i = \mathbf{p}_f - \mathbf{q}_{\text{tot}}. \tag{43}$$

This is the direct local-current analogue of a two-current insertion into the quark line. It is intended as a diagnostic building block for current-current and hadronic-tensor style measurements. As with the first-order response, tau windows are summed at the inserted-source level before inversion; the code does not save per-τ response-propagator caches.

9.5 Diagnostic application

The current first-order Perlmutter diagnostic application is

`application/EMFF_pion_background_response/perlmutter/`.

The default run uses the S8T32 test gauge and compares

$$C_{\mathrm{resp}}(t_{\mathrm{sep}}) \quad \text{against} \quad \sum_{\tau} C_3(t_{\mathrm{sep}}, \tau) \tag{44}$$

for $t_{\mathrm{sep}} = 2$, $\Gamma = \gamma_T$, and zero external momentum. The upgraded diagnostic can also run several t_{sep}, several momentum transfers, and several current gamma matrices in a single job. The HDF5 output stores one record for each $(\Gamma, \mathbf{q}, t_{\mathrm{sep}}, W_\tau)$.

10 Implemented Upgrade Path

1. **Nonzero-q validation.** The response schema stores

   $$\mathbf{p}_i = \mathbf{p}_f - \mathbf{q},$$

   for every record. The phase test covers both ordinary and Breit-frame examples, including

   $$\mathbf{p}_f = (0,0,1), \quad \mathbf{q} = (0,0,1), \quad \mathbf{p}_i = (0,0,0).$$

   and

   $$\mathbf{p}_f = (0,0,1), \quad \mathbf{q} = (0,0,2), \quad \mathbf{p}_i = (0,0,-1).$$

2. **Multi-t_{sep}, C_2, and summed-ratio output.** The application accepts a list of source-sink separations and saves

   $$C_2(t_{\mathrm{sep}}), \quad C_{\mathrm{resp}}(t_{\mathrm{sep}}), \quad R_{\mathrm{sum}}(t_{\mathrm{sep}}).$$

   The saved ratios are

   $$R^{\mathrm{resp}}_{\mathrm{sum}}(t_{\mathrm{sep}}) = \frac{C_{\mathrm{resp}}(t_{\mathrm{sep}})}{C_2(t_{\mathrm{sep}})}, \qquad R^{\mathrm{explicit}}_{\mathrm{sum}}(t_{\mathrm{sep}}) = \frac{\sum_{\tau \in W_\tau} C_3(t_{\mathrm{sep}}, \tau)}{C_2(t_{\mathrm{sep}})}. \tag{45}$$

3. **Restricted tau windows.** The supported choices are all time slices, source-to-sink, open source-to-sink, a symmetric restricted window, and an explicit integer range:

   $$W_\tau = \{\tau_{\min}, \ldots, t_{\mathrm{sep}} - \tau_{\min}\}.$$

   For restricted windows, the inserted source is summed before inversion.

4. **Multi-current gamma response suite.** The application accepts multiple gamma labels from the standard pion gamma list. The primary EMFF current is still V_4, labeled `T`, but scalar, pseudoscalar, vector, axial, and tensor bilinears can be used as current-response diagnostics without changing the first-order contraction.

5. **HDF5 schema upgrade.** Version-2 output stores file-level kinematics and one `results/record_XXXX` group per diagnostic. Each record includes $\mathbf{p}_f$, $\mathbf{q}$, inferred $\mathbf{p}_i$, t_{sep}, current gamma, source/sink gamma labels, the tau-window definition, C_2, explicit summed C_3, response C_2-like data, ratios, and the response-explicit difference. For analysis convenience, the same per-record scalars are also collected in `summary/` datasets:

   `relative_difference`, `response_R_sum`, `explicit_R_sum`,

   together with $\mathbf{p}_f$, $\mathbf{q}$, $\mathbf{p}_i$, t_{sep}, gamma labels, and window labels. Analysis scripts can therefore scan the diagnostics without opening every `results/record_XXXX` group.

6. **Optional GPU regression tests.** The Perlmutter application remains a small S8T32 smoke workflow. The test suite contains CPU-only schema, phase, and algebra checks, while the GPU smoke run can be launched when a QUDA-enabled node is available.

Explicit non-goal. This implementation does not save a per-τ response-propagator cache. If arbitrary post-hoc tau windows become necessary, that should be considered as a separate design discussion rather than silently added to this diagnostic.

11 Summary

The present pion current background-response code is a linearized Feynman-Hellmann diagnostic:

$$S_{\mathrm{resp}} = D^{-1}\Gamma\Phi_{\mathbf{q}}S. \tag{46}$$

It does not compute finite-λ energies. Instead, it computes $\partial C_\lambda/\partial\lambda|_{\lambda=0}$, which equals a summed three-point function up to the convention-dependent sign. The energy-shift derivative is then encoded in the slope of the response ratio C_{resp}/C_2 as a function of t_{sep}.